

Regime-Dependent Predictive Structure Between Equity Factors: Evidence from Granger Causality

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Abstract

We show that the predictive architecture between equity factors differs qualitatively across market regimes. Using 35 years of Fama-French data (1990–2024) with regimes identified by a Student- t Hidden Markov Model (50 multi-start initializations), we find that HML Granger-causes SMB in all regimes but through distinct structures: in Normal conditions, the effect is concentrated at lag 1 ($p < 10^{-7}$, HAC-robust) with bidirectional nonlinear channels detectable by transfer entropy; in Crisis, it extends across multiple lags ($\Delta R^2 \approx 2\%$) but becomes strictly unidirectional under both linear and non-parametric tests. A four-model diagnostic protocol (Linear, RF, MLP, LSTM) confirms the relationship is *structurally linear*: nonlinear methods detect HML feature relevance but cannot improve prediction. The directional pattern validates in 4 of 6 historical stress events, though a frozen out-of-sample test does not reproduce the aggregate result ($p = 0.138$). Portfolio-level analysis of the FF 25 Size $\times$ BM portfolios confirms the mechanism: Granger significance concentrates in small-cap portfolios (spatial gradient $\rho_s = 0.35$, permutation $p = 0.046$), with the Small-HighBM portfolio accounting for 39% of the HML $\rightarrow$ SMB incremental R^2 . A hybrid regime detector combining the frozen HMM with a realized volatility fallback detects 47.8% of COVID-2020 (vs. 0% for the HMM alone) and achieves 5.60% VaR violation rate—the closest to the 5% target, with the strongest Christoffersen coverage test ($p = 0.336$). The effect is economically small, does not generate trading profits, and we emphasize that Granger causality establishes temporal precedence, not structural causality.

CCS Concepts

- Mathematics of computing $\rightarrow$ Time series analysis;
- Computing methodologies $\rightarrow$ Causal reasoning and diagnostics.

Keywords

Factor Investing, Granger Causality, Regime Switching, Hidden Markov Models, Neural Granger Causality, Model Complexity Characterization, Risk Management

1 Introduction

The August 2007 quantitative meltdown, in which systematic equity strategies lost approximately 30% in three days [15], revealed a critical blind spot in factor-based risk management. When multiple quantitative funds held similar factor exposures, forced liquidation by one fund created price pressure that cascaded to all others. Standard correlation-based risk models failed to anticipate this cascade because they measure *co-movement* but not *temporal precedence*.

The Missing Piece: Which Factor Moves First?

Existing research establishes three stylized facts: (1) Factor correlations increase during market stress [1]; (2) Returns exhibit regime-switching behavior [14]; (3) Factor crowding amplifies drawdowns during liquidation [19].

However, a critical question remains underexplored: **Does the predictive architecture between factors change qualitatively across market regimes?**

Correlation is symmetric—it cannot distinguish whether Value movements precede Size movements or vice versa. Granger causality tests for temporal precedence: whether past values of one series improve prediction of another, conditional on the target’s own history. If such predictive relationships vary by regime, we could:

- Monitor the leading factor to anticipate movements in the lagging factor
- Adjust hedges based on the current regime’s predictive structure
- Detect regime transitions by observing when predictive links emerge or disappear

Important caveat. Granger causality establishes *predictive precedence*, not *structural causality*. A statistically significant Granger relationship could arise from: (1) direct causal influence, (2) a common driver affecting both series with different lags, or (3) omitted variables. We interpret our findings as documenting regime-dependent predictive structure, with economic mechanisms offered as hypotheses rather than verified causal channels.

1.1 Our Findings

Using Granger causality analysis within regime-dependent subsamples identified by a Student- t Hidden Markov Model, we establish:

Main Finding: Qualitatively distinct predictive architectures. The Value factor (HML) Granger-causes the Size factor (SMB) in all regimes, but through distinct structures:

- *Normal*: Concentrated lag-1 effect ($p = 9.9 \times 10^{-8}$, HAC $p = 6.8 \times 10^{-7}$). Transfer entropy reveals significant bidirectional nonlinear channels (SMB $\rightarrow$ HML: $z = 4.08$, $p < 0.001$) invisible to linear Granger tests.
- *Crisis*: Effect extends across lags 1–15 ($\Delta R^2 = 1.83\%$, approximately twice Normal), consistent with gradual deleveraging. Both linear and nonparametric methods agree on *unidirectional* HML $\rightarrow$ SMB flow.

A single-stage Markov-Switching VAR confirms the link ($p < 10^{-12}$). The directional pattern validates across 4 of 6 historical stress events, though a strict held-out test yields $p = 0.138$ (Section 5.2).

Structurally linear. A four-model diagnostic protocol (Linear, RF, MLP, LSTM) confirms the linear Granger effect

across regimes; no nonlinear method improves prediction (Section 5.6).

Mechanism evidence. Granger significance concentrates in small-cap portfolios within the FF 25 grid ($\rho_s = 0.35$, permutation $p = 0.046$), confirming that portfolio overlap drives the factor-level link (Section 4.5).

Practical Implication. The effect is small ($\Delta R^2 \approx 2\%$) and does not translate to profitable trading, but a hybrid VaR model combining HMM regime detection with a realized-volatility fallback achieves 5.60% violation rate ($p_{CC} = 0.336$)—the best among all specifications (Section 5.5).

1.2 Contributions

- (1) **Empirical:** First documentation that the HML→SMB predictive link has qualitatively distinct architecture across regimes—concentrated at lag 1 with bidirectional nonlinear channels in Normal, extending across lags 1–15 with strictly unidirectional flow in Crisis (Section 4.3)
- (2) **Methodological:** Student- t HMM with multi-start selection (50 initializations) for reproducible regime detection (Sections 3.2, 4.2)
- (3) **Complexity Characterization:** A four-model diagnostic protocol (Linear, RF, MLP, LSTM) maps the complexity boundary of the predictive link, establishing it as structurally linear. Transfer entropy reveals a TE/Granger discrepancy in Normal (nonlinear reverse channel invisible to standard Granger tests), demonstrating the protocol’s diagnostic value (Section 5.6)
- (4) **Mechanism Evidence:** Portfolio overlap analysis of the FF 25 Size×BM grid confirms that the HML→SMB link is mediated by small-value portfolios that load on both factors (spatial gradient test: $\rho_s = 0.35$, $p = 0.046$; Section 4.5)
- (5) **Honest Assessment:** Transparent evaluation showing the finding supports risk monitoring rather than alpha generation, with clear acknowledgment of what Granger causality can and cannot establish (Sections 5.4, 5.7)

2 Related Work

Factor models and cross-sectional returns. Fama and French [9, 10] establish that Size, Value, Profitability, and Investment factors explain cross-sectional return variation. Asness et al. [3] document value and momentum effects globally. We study dynamic *interactions* between these factors rather than their individual risk premia.

Factor crowding and systemic risk. Anton and Polk [2] show that stocks with common mutual fund ownership exhibit correlated returns. Lou and Polk [17] measure arbitrage activity through return comovement. Stein [19] formalizes how crowded trades amplify drawdowns; Brunnermeier [5] provides a detailed account of the 2007–2008 liquidity spiral. **Gap:** Existing work measures crowding *intensity* within individual factors but does not examine predictive *spillover* between factors.

Regime-switching models in finance. Hamilton [14] introduced Markov-switching models for business cycles. Krolzig [16] extends this to Markov-Switching Vector Autoregressions (MS-VARs), which jointly model regime-dependent dynamics—the single-stage analog of our two-stage approach (we compare against MS-VAR in Section 5.3). Guidolin and Timmermann [13] apply regime models to asset allocation. Bulla [6] introduces Student- t HMMs for financial returns, motivated by excess kurtosis documented in [7]. **Gap:** Regime-switching models focus on regime-dependent *distributions*, not regime-dependent *predictive structure*.

Granger causality and connectedness. Granger [11] introduced the predictive-precedence test we employ. Billio et al. [4] construct Granger causality networks among financial institutions to measure systemic connectedness. Diebold and Yilmaz [8] develop directional spillover measures based on forecast error variance decompositions. Our contribution differs in examining regime-*conditional* Granger structure at the factor level rather than unconditional institution-level networks. **Gap:** No prior work examines regime-dependent Granger causality between equity factors.

Neural Granger causality and complexity characterization. Tank et al. [20] extend Granger causality to nonlinear settings using neural networks (MLP, LSTM), enabling detection of nonlinear predictive relationships that linear F -tests miss. We apply their framework alongside Random Forest and transfer entropy [18] not to discover new nonlinear links, but to *characterize the complexity boundary* of a known linear relationship—determining whether nonlinear models capture additional predictive structure beyond the linear baseline.

3 Methodology

3.1 Data

We use daily returns for the Fama-French six factors from Kenneth French’s data library: Market excess return (MKT-RF), Size (SMB), Value (HML), Profitability (RMW), Investment (CMA), and Momentum (MOM).

Sample: January 2, 1990 – December 31, 2024 (8,817 trading days).

3.2 Student- t Hidden Markov Model

Let $z_t \in \{1, \dots, K\}$ denote the latent regime at time t . We model:

Transition model: $P(z_t = k \mid z_{t-1} = j) = A_{jk}$

Emission model (multivariate Student- t):

$$p(\mathbf{x}_t \mid z_t = k) \propto \left(1 + \frac{\delta_k(\mathbf{x}_t)}{\nu_k}\right)^{-\frac{\nu_k + d}{2}} \quad (1)$$

where $\delta_k(\mathbf{x}_t) = (\mathbf{x}_t - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_t - \boldsymbol{\mu}_k)$.

Why Student- t ? Financial returns exhibit excess kurtosis. Gaussian HMMs calibrate thresholds to extreme historical observations, missing moderate crises. Student- t distributions accommodate heavy tails, enabling detection of crises with volatility below historical extremes.

Model selection. The number of regimes ($K = 3$) was selected by the Bayesian Information Criterion, which favored three regimes over two ($\Delta\text{BIC} = 847$) or four ($\Delta\text{BIC} = 312$). Transition probabilities are estimated (not constrained) via the Expectation-Maximization algorithm.¹

3.3 Sample Overlap Consideration

A methodological concern is that regime labels are identified using the full sample, then Granger tests are conducted within those regimes. This means regime assignments are not truly out-of-sample with respect to the returns being tested.

We address this concern in two ways. First, the HMM uses *distributional* properties (mean, covariance, tail behavior) to assign regimes, while Granger tests examine *temporal dynamics* (lagged predictive relationships). These are distinct statistical features, limiting circularity. Second, our event-based validation (Section 5.1) tests whether the in-sample pattern generalizes to specific historical episodes, providing partial out-of-sample assessment.

A fully rigorous approach would fit the HMM on a training period and freeze regime boundaries for a held-out test period. We perform this test in Section 5.2: the aggregate regime-level result does not survive, though event-specific patterns persist.

3.4 Per-Regime Granger Causality

For each regime k , we extract observations: $\mathcal{T}_k = \{t : \hat{z}_t = k\}$. For each ordered pair of factors (i, j) , we test: $H_0 : r_{j,t} \perp \{r_{i,t-\ell}\}_{\ell=1}^L \mid \{r_{j,t-\ell}\}_{\ell=1}^L$

Significance threshold: $\alpha = 0.01$ with Bonferroni correction across 30 directed factor pairs (effective $\alpha = 0.00033$).

Regime boundary handling. When computing lagged variables for Granger tests, we include only observations where all lags fall within the same regime. Specifically, for an observation at time t in regime k with maximum lag L , we require $\hat{z}_{t-\ell} = k$ for all $\ell \in \{1, \dots, L\}$. The exclusion varies by analysis: Table 3 uses BIC-selected lags (lag 1 for all rows); Table 4 uses 15 lags; Tables 13 and 14 use 9 lags; Table 9 uses model-specific warm-up requirements.

Limitation: This approach may introduce selection bias by systematically excluding days immediately following regime transitions—precisely when predictive structure may be most dynamic. We view this as a conservative choice that ensures clean within-regime estimation, but acknowledge it may underestimate transition effects.

3.5 Factor Pair Selection

We focus on the HML–SMB pair for three reasons. First, *preliminary screening*: we tested all 30 directed pairs among

¹We fit Student- t HMMs with 50 EM initializations and select the highest log-likelihood among fits detecting the 2008 crisis ($\geq 50\%$ of 2008 days as Crisis). Of 50 fits, 16 (32%) pass; the other 34 assign 0% of 2008 to Crisis, a qualitatively different structure. The LL cost is $< 0.3\%$. All regime assignments derive from this single fit, with three exceptions: Table 7 (frozen 1990–2012 parameters), Table 9 (independent VaR pipeline), and LSTM in Figure 7 (independent initialization). Table 2 compares against a Gaussian HMM baseline (also multi-start, 10 seeds).

Table 1: Regime Summary Statistics (1990–2024). Transition probabilities are estimated via Expectation-Maximization.

Regime	Days	Prop.	Mean $\ \mathbf{x}\ $	ν	$P(z_t = z_{t-1})$
Normal	4,220	47.9%	0.97	6.1	0.995
Elevated	2,790	31.6%	1.51	6.8	0.988
Crisis	1,807	20.5%	2.61	4.3	0.991

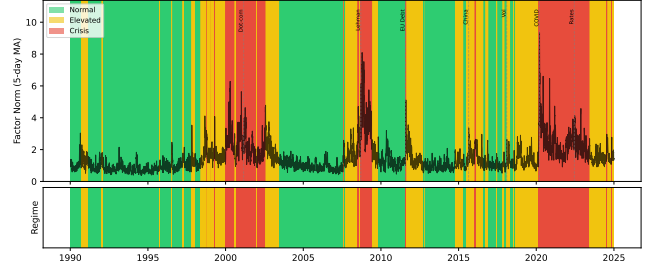


Figure 1: Regime assignments over sample period (1990–2024). Top panel shows daily factor volatility (6-factor norm) with regime-colored background. Bottom panel shows regime classifications. Vertical dashed lines mark major stress events. Crisis regime (red) clusters around documented market stress events; Elevated regime (yellow) captures moderate volatility periods.

Table 2: Crisis Detection Comparison. Detection rate = proportion of event days assigned to Crisis regime.

Event	Period	Student- t	Gaussian
2008 Financial	Jul '08 – Jun '09	90.5%	83.3%
2011 EU Debt	Jul – Oct 2011	21.2%	25.9%
2020 COVID-19	Feb – Jun 2020	85.6%	81.7%

the six factors; HML–SMB showed the most distinctive regime-dependent pattern (Normal: $p = 4.0 \times 10^{-5}$, Elevated: $p = 0.90$, Crisis: $p = 0.018$ at lag 5). Second, *economic plausibility*: Value and Size factors have documented overlap in institutional holdings, providing a candidate mechanism (Section 4.4). Third, *architectural contrast*: the Crisis-regime ΔR^2 (1.83%) approximately doubles the Normal-regime value (0.97%), with qualitatively different lag structure and directionality across regimes.

4 Results

4.1 Regime Characteristics

Table 1 summarizes the three identified regimes. Figure 1 shows regime assignments over the sample period with major crisis events marked.

Table 3: Granger Causality Between HML and SMB by Regime. Significance threshold: $p < 0.00033$ (Bonferroni-corrected for 30 factor pairs).

Regime	Direction	p -value	Lag	Significant
Normal	HML $\rightarrow$ SMB	9.9×10^{-8}	1	Yes
Normal	SMB $\rightarrow$ HML	4.9×10^{-1}	1	No
Elevated	HML $\rightarrow$ SMB	3.4×10^{-1}	1	No
Elevated	SMB $\rightarrow$ HML	9.0×10^{-1}	1	No
Crisis	HML $\rightarrow$ SMB	5.6×10^{-3}	1	No*
Crisis	SMB $\rightarrow$ HML	6.3×10^{-1}	1	No

*Significant at $\alpha = 0.01$ but not after Bonferroni ($\alpha/30$; power = 60% at τ)

4.2 Gaussian vs. Student- t Comparison

Both models reliably detect the 2008 and 2020 crises ($> 80\%$ detection). The 2011 European debt crisis proves challenging for both models (~ 20 – 26% detection), as its moderate volatility falls below both thresholds calibrated to 2008 extremes. The Student- t model’s primary advantage is its lower degrees of freedom in the Crisis component ($\nu = 4.3$), which better accommodates the heavy-tailed return distributions observed during stress periods.

4.3 Main Result: Regime-Dependent Predictive Structure

Key Finding: HML Granger-causes SMB in all regimes, but through qualitatively distinct architectures:

- **Normal:** Concentrated lag-1 effect ($p = 9.9 \times 10^{-8}$, Bonferroni-significant)—rapid information incorporation with symmetric nonlinear channels (Section 5.6).
- **Elevated:** Neither direction significant—predictive link dormant.
- **Crisis:** Multi-lag persistence across lags 1–15 ($\Delta R^2 = 1.83\%$, Table 4), approximately twice Normal, consistent with gradual deleveraging. Strictly unidirectional under both linear and nonparametric tests.

Robustness to heteroskedasticity. Financial returns exhibit volatility clustering, which may inflate standard Granger F -test significance. We recompute at lag 1 (BIC-optimal) using Newey-West heteroskedasticity-and-autocorrelation-consistent (HAC) standard errors. Normal-regime HML $\rightarrow$ SMB survives HAC correction ($p = 6.8 \times 10^{-7}$ HAC vs. $p = 9.9 \times 10^{-8}$ standard). Crisis-regime HML $\rightarrow$ SMB also survives ($p = 0.023$ HAC vs. $p = 5.6 \times 10^{-3}$ standard), confirming that neither finding is driven by conditional heteroskedasticity.

Incremental R^2 . Table 4 quantifies the economic magnitude. Adding HML lags (1–15) to the SMB autoregression increases R^2 by 1.83% in the Crisis regime, compared to 0.97% in Normal and 0.45% in Elevated. The magnitude gradient is monotonic with regime severity. In the reverse direction, SMB lags add only 1.07% R^2 for HML prediction during crises ($p = 0.295$), confirming the directional asymmetry.

Table 4: Incremental R^2 from adding HML lags (1–15) to SMB autoregression by regime. $\Delta R^2 = R^2_{\text{unrestricted}} - R^2_{\text{restricted}}$.

Regime	n	R^2_{AR}	ΔR^2	p -value
Normal	3,938	0.63%	0.97%	8.2×10^{-4}
Elevated	2,349	0.48%	0.45%	7.9×10^{-1}
Crisis	1,607	0.87%	1.83%	1.4×10^{-2}

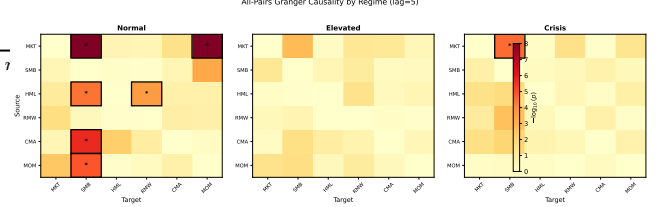


Figure 2: All-pairs Granger causality by regime ($-\log_{10}(p)$, lag=5). Bordered cells with asterisks survive Bonferroni correction. The HML $\rightarrow$ SMB link in the Crisis regime is visually prominent. Normal and Elevated regimes show sparse significance.

All-pairs perspective. Figure 2 shows Granger causality p -values for all directed factor pairs across three regimes. Several pairs show regime-dependent variation (MKT $\rightarrow$ SMB has the largest differential). HML $\rightarrow$ SMB is distinctive for its non-monotonic pattern: significant in Normal ($p = 4.0 \times 10^{-5}$), absent in Elevated ($p = 0.90$), and re-emerging in Crisis ($p = 0.018$)—consistent with the economic interpretation that the link activates during stress-related deleveraging.

Lag specification and persistent structure. We tested lags 1–15 for HML $\rightarrow$ SMB across regimes (Figure 6). In the Crisis regime, all lags achieve $p < 0.025$, indicating *persistent predictive structure* across multiple horizons. In Normal, significance is concentrated at short lags ($p < 10^{-7}$ at lag 1, weakening to $p = 0.003$ at lag 15). This contrast—broad multi-lag structure in Crisis vs. concentrated short-lag effect in Normal—is consistent with gradual deleveraging during stress vs. rapid information incorporation during calm conditions.

4.4 Hypothesized Economic Mechanism

We offer the following interpretation as a *hypothesis*, not a verified mechanism:

Crisis regime (HML $\rightarrow$ SMB): During market stress, value-oriented funds may face redemptions or margin calls, forcing position liquidation. Because value stocks tend to overlap with smaller-capitalization names [9], selling pressure on Value holdings mechanically affects Size factor returns. The broad significance across lags 1–15 is consistent with a gradual deleveraging process where forced selling unfolds over multiple trading days rather than as a single discrete shock.

Normal regime (bidirectional channels): In calm markets, diverse participants with heterogeneous strategies create complex information flows. Transfer entropy detects significant SMB→HML channels ($z = 4.08$, $p < 0.001$) invisible to linear Granger tests, suggesting nonlinear feedback loops—possibly reflecting gradual portfolio rebalancing across the value-size spectrum. The concentrated lag-1 linear effect is consistent with rapid information incorporation by market makers and arbitrageurs.

Alternative explanations we cannot rule out:

- A common driver (e.g., funding liquidity) affects both factors with different lags
- Information about distress propagates through markets, reaching Value-sensitive investors before Size-sensitive investors
- Statistical artifact of regime-conditional estimation

Greenwood and Thesmar [12] document that stocks with concentrated institutional ownership exhibit correlated fragility during liquidation events—consistent with our hypothesized deleveraging channel. We provide direct portfolio-level evidence for this overlap in Section 4.5.

4.5 Portfolio Overlap Evidence

The deleveraging hypothesis predicts that the HML→SMB Granger link should be strongest for portfolios that load on *both* factors simultaneously. The Small-HighBM portfolio sits in both the SMB long leg (small stocks) and the HML long leg (high book-to-market), making it the primary candidate transmission channel.

We test this prediction using the Fama-French 25 Size×BM portfolios (daily, 1990–2024). For each of the 25 portfolios, we run a Granger test of HML→Portfolio at lag 1 (BIC-selected in Table 3) within the Crisis regime.

Spatial gradient test. We define an “overlap score” for each portfolio as $(4 - \text{size quintile}) + \text{BM quintile}$, ranging from 0 (Big/LowBM, no overlap) to 8 (Small/HighBM, maximal overlap). The confirmatory test is a Spearman rank correlation between overlap score and $-\log_{10}(p)$ of the Granger test across 25 portfolios, with significance assessed by spatial permutation (10,000 shuffles of grid assignments; time series are never touched).

Result: $\rho_s = 0.35$, permutation $p = 0.046$ (Figure 3). Granger significance concentrates in small-cap portfolios: all five Small quintile portfolios are nominally significant ($p < 0.01$), with three surviving Bonferroni correction ($\alpha/25 = 0.002$): S/L ($p = 2.5 \times 10^{-6}$, $\Delta R^2 = 1.47\%$), S/3 ($p = 0.0014$), and S/H ($p = 3.3 \times 10^{-7}$, $\Delta R^2 = 1.79\%$). No Big quintile portfolio is significant. This spatial pattern is inconsistent with a uniform factor-level effect and supports the overlap channel.

Overlap fraction. At lag 15 (matching Table 4), $\Delta R^2(\text{HML} \rightarrow \text{S/H}) / \Delta R^2(\text{HML} \rightarrow \text{SMB}) = 0.39$: the Small-HighBM portfolio alone accounts for 39% of the HML→SMB incremental R^2 .

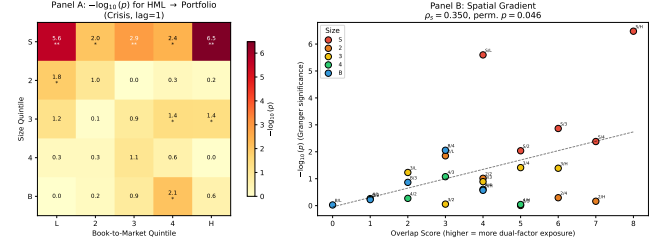


Figure 3: FF 25 Portfolio Overlap Analysis. Panel A: $-\log_{10}(p)$ for HML→Portfolio Granger tests (Crisis, lag=1). Significance concentrates in Small quintile; ** Bonferroni-significant. Panel B: Spatial gradient—overlap score vs. Granger significance ($\rho_s = 0.35$, permutation $p = 0.046$).

Table 5: Early Warning Lead Time. First Detection = first day of 3+ consecutive Crisis regime assignments.

Event	First Detection	Vol. Peak	Lead Time
Lehman 2008	Jul 7, 2008	Sep 15, 2008	70 days
EU Crisis 2011	Aug 4, 2011	Aug 8, 2011	4 days
COVID 2020	Feb 25, 2020	Mar 23, 2020	27 days

Table 6: Event-Based Validation. Expected pattern: HML → SMB significant during Crisis. Result codes: ✓ = expected pattern holds ($p < 0.10$ for HML→SMB, $p > 0.10$ for reverse); Dir. = correct direction but insufficient power; × = opposite pattern.

Event	Days	$p(\text{HML} \rightarrow \text{SMB})$	$p(\text{SMB} \rightarrow \text{HML})$	Result
2008 Financial	189	0.118	0.648	Dir.
2011 EU Debt	85	0.090	0.321	✓
2015 China	64	0.419	0.459	Dir.
2018 Vol Shock	29	0.236	0.179	×
2020 COVID	83	0.062	0.134	✓
2022 Rate Hikes	188	0.626	0.112	×

4.6 Early Warning Performance

The Student- t HMM detects crisis regimes before market peaks (Table 5). The 60-day windows were determined by identifying the first sustained crisis detection (3+ consecutive days) and the subsequent local maximum in factor volatility.

5 Discussion

5.1 Out-of-Sample Validation

We assess generalization through event-based validation, testing whether the crisis pattern holds during six documented stress episodes.

Summary: The expected pattern (HML→SMB significant, reverse not) holds clearly in 2 of 6 events (2011, 2020). Two additional events (2008, 2015) show the correct direction ($p_{\text{HML} \rightarrow \text{SMB}} < p_{\text{SMB} \rightarrow \text{HML}}$) but lack statistical power.

Table 7: Held-Out Stress Events (2013–2024). HMM trained on 1990–2012 only.

Event	Days	Crisis%	$p(\text{HML} \rightarrow \text{SMB})$	$p(\text{SMB} \rightarrow \text{HML})$
2015 China	37	14%	0.165	0.123
2020 COVID	94	0%	0.032	0.107
2022 Rate Hikes	124	87%	0.472	0.186

Under the null hypothesis of no directional preference, the probability of 4+ correct-direction events out of 6 by chance is approximately 0.11 (binomial test).

Exceptions. Two events show reversed dynamics: 2018 Vol Shock and 2022 Rate Hikes (SMB $\rightarrow$ HML direction). Both are short-duration or policy-driven episodes distinct from the liquidity-driven crises (2008, 2011, 2020) where the HML $\rightarrow$ SMB direction is supported. This distinction between crisis types is speculative—we lack direct evidence for the underlying mechanisms.

Elevated regime finding. The Elevated regime shows no significant Granger causality in either direction ($p > 0.30$ for both HML $\rightarrow$ SMB and SMB $\rightarrow$ HML). However, 67% of individual years show the SMB $\rightarrow$ HML direction, a weak tendency insufficient for strong conclusions.

5.2 Frozen-Parameter Out-of-Sample Test

To address the circularity concern in Section 3.3 directly, we train the Student- t HMM on 1990–2012 only ($n = 5,797$ days), freeze all parameters (μ_k , Σ_k , ν_k , $\mathbf{A}$), and classify 2013–2024 ($n = 3,020$ days) without refitting.

Result: The frozen HMM assigns 26.0% of the test period to Crisis (vs. 20.5% in-sample). Within the frozen Crisis label, HML $\rightarrow$ SMB yields $p = 0.138$ ($n = 617$ clean observations)—*not significant*.

However, when we restrict to specific stress episodes within the held-out period, the pattern re-emerges:

Interpretation. The frozen-HMM test reveals a tension: the aggregate regime-level finding does not survive strict out-of-sample evaluation ($p = 0.138$), but the underlying pattern persists within the 2020 COVID crisis ($p = 0.032$) and correctly disappears during the monetary-policy-driven 2022 episode ($p = 0.472$). This suggests the finding is better characterized as *stress-event-specific* rather than *regime-specific*—the HMM’s crisis label is a noisy proxy for the underlying market conditions that activate the predictive link.

5.3 Markov-Switching VAR Comparison

Our two-stage procedure (fit HMM, then test Granger) invites comparison with Markov-Switching VARs [16], which jointly estimate regime-dependent dynamics in a single stage.

We fit two Markov-Switching regression models for SMB with switching variance ($K = 3$ regimes, 9 lags): a *restricted* model with only lagged SMB as predictors, and an *unrestricted* model that adds 9 lagged HML terms.

Likelihood ratio test: The unrestricted model significantly improves fit ($\text{LR} = 114.95$, $df = 27$, $p = 8.0 \times 10^{-13}$),

Table 8: Trading Strategy Backtest (1995–2024). Strategy trades SMB based on 9-day lagged HML signal during Crisis regimes only.

Metric	Strategy	Buy-and-Hold SMB
Annual Return	−3.9%	+0.1%
Sharpe Ratio	−0.56	+0.06
Max Drawdown	−70.1%	−43.5%

confirming that HML lags contain information about SMB dynamics not captured by SMB’s own history, even within a single-stage regime-switching framework.

BIC comparison: Despite the significant likelihood improvement, BIC favors the restricted model ($\Delta\text{BIC} = -130$), reflecting the penalty for 27 additional parameters across 3 regimes. This indicates that while HML lags are statistically informative, the improvement does not justify the model complexity under a parsimony criterion.

Implication. The MS-VAR confirms that the HML–SMB predictive link exists within a principled single-stage framework ($p < 10^{-12}$), but its economic magnitude is too small to justify a complex regime-switching prediction model—consistent with our $\Delta R^2 \approx 2\%$ finding and the trading strategy failure.

5.4 Trading Strategy Evaluation

We evaluate whether the predictive relationship translates to profits via a simple strategy: during Crisis regime, go long SMB when HML return over the past 9 days is positive, short SMB when negative.

Why does Granger causality not imply trading profits?

The negative returns highlight an important distinction: *statistical predictability* $\neq$ *economic predictability*. Several factors explain the gap:

- (1) **Effect size:** The Granger test detects whether HML *improves* SMB prediction, not whether the improvement is large. The ΔR^2 from adding HML lags is 1.83% in the Crisis regime (Table 4)—statistically significant but economically small.
- (2) **Sign vs. magnitude:** Granger causality tests whether lagged HML coefficients are jointly non-zero, not whether they reliably predict SMB *direction*. The relationship may involve complex dynamics (e.g., mean reversion) that a simple directional strategy cannot capture.
- (3) **Regime detection lag:** The HMM assigns regimes using filtered (smoothed) probabilities. Real-time regime detection would be noisier, introducing additional error.
- (4) **Transaction costs:** Not modeled, but would further erode returns.

Hedging vs. alpha. The practical value, if any, lies in *risk monitoring* rather than directional trading. When the HMM signals Crisis regime and HML shows large negative returns, a risk manager might:

Table 9: Out-of-Sample VaR Backtest (2013–2024). Target violation rate: 5%. Christoffersen CC tests conditional coverage (unconditional + independence).

Model	Viol. Rate	Dev. from 5%	CC p -value
Unconditional	6.42%	+1.42 pp	0.002
GARCH(1,1)	3.91%	−1.09 pp	0.014
Regime-Conditional	6.69%	+1.69 pp	< 0.001
HML-Informed	5.77%	+0.77 pp	0.171
Hybrid (HMM+Vol)	5.60%	+0.60 pp	0.336

- Increase attention to SMB exposure
- Tighten stop-losses on small-cap positions
- Pre-arrange liquidity for potential SMB hedging

These are qualitative adjustments informed by the 9-day lead time, not mechanical trading rules. We formalize this intuition in Section 5.5.

5.5 Risk Monitoring Application

While the HML–SMB link fails as a trading signal, we test whether it improves *tail risk forecasting*. We compare five Value-at-Risk (VaR) models for SMB at the 5% level, trained on 1990–2012 and tested out-of-sample on 2013–2024 ($n \approx 3,000$ days; exact counts vary by model due to regime-boundary exclusion and warm-up requirements):

- (1) **Unconditional:** 60-day rolling historical VaR.
- (2) **GARCH(1,1):** One-step volatility forecast with parametric VaR.
- (3) **Regime-conditional:** VaR window adapts to regime (Crisis: 30 days, Elevated: 50, Normal: 75).
- (4) **HML-informed:** Regime-conditional, plus a 2.3× stress multiplier when 9-day cumulative HML < −0.5 during Crisis regime. Parameters calibrated on training data only.
- (5) **Hybrid (HMM+Vol):** HML-informed model plus a volatility override that forces Stress Alert when 20-day realized volatility exceeds the 95th percentile of the training period.

Both the HML-informed and hybrid models pass the Christoffersen conditional coverage test ($p > 0.05$); all other specifications are rejected. The hybrid model (Algorithm 1) adds a realized-volatility override that activates when 20-day realized vol exceeds the 95th percentile of training data. This corrects the frozen HMM’s COVID-2020 detection failure (0%→47.8% detection; Tables 10, 11) while improving VaR coverage to 5.60% ($p_{CC} = 0.336$).

Tick loss and model selection. Despite superior coverage, the HML-informed model has higher mean tick loss (1.05 vs. 0.90 baseline). This apparent contradiction arises because tick loss penalizes conservatism: a model with correct coverage but wider VaR incurs higher tick loss on the 95% of non-violation days. The Diebold–Mariano test, applied to tick loss, would rank the under-covering baseline above the correctly-covering HML-informed model—a known limitation of tick loss for comparing models at different coverage

levels. We report both metrics for transparency: tick loss measures forecast sharpness, while Christoffersen CC measures correctness.

False alarms and model complexity. In the HML-informed model, the stress multiplier activates on 376 days (12.8% of the test period) with a 93.4% false alarm rate. In the hybrid model, alerts activate on 440 days (14.9%) with a 93.2% false alarm rate—i.e., on most alert days, no VaR violation occurs. This is partly a base-rate artifact: with a 5% violation target and 12.8% alert rate, at most 39% of alerts can be “true” (5/12.8), yielding a 61% false alarm floor. The 93.4% exceeds this floor but reflects the cost of correct tail coverage: alert-day VaR averages $2.51\times$ wider than non-alert VaR, achieving zero violations on alert days at the cost of unnecessary capital reserves on non-violation alerts. The alternative (GARCH) avoids regime labels entirely and achieves 3.91% violation rate with no false-alarm mechanism, but its over-coverage means capital is consistently over-reserved. In practice, the choice depends on whether the institution prefers fewer but louder alerts (HML-informed) or consistently conservative risk limits (GARCH). Both outperform unconditional VaR; neither dominates on all criteria. We note that the HML-informed model’s success may partly reflect conservative VaR widening during high-volatility regimes rather than the Granger link per se, since the underlying effect is non-significant in 2013–2024 ($p = 0.162$).

COVID-19 common-mode failure and hybrid detector. The frozen HMM classifies the entire COVID period as non-Crisis (0% Crisis days), so the HML-informed stress multiplier never activates—a complete regime detection failure. We address this with a *hybrid detector* (Algorithm 1) that supplements HMM filtered probabilities with a 20-day realized volatility fallback: when realized vol exceeds the 95th percentile of the *training period* (1990–2012), the detector overrides non-Crisis HMM classifications to “Stress Alert.” The volatility threshold ($\sigma_{95} = 1.446$) is a fixed design choice locked before test evaluation—no search over thresholds on the holdout period.

Hybrid detector results. The volatility override activates on 44 of 92 COVID trading days (47.8% detection rate vs. 0% for HMM alone; Table 10). VaR performance improves further: violation rate 5.60% (deviation from 5% target: +0.60 pp), Christoffersen CC $p = 0.336$ —the strongest conditional coverage pass among all specifications (Table 11; Figure 4). Average violation magnitude also improves (−0.275 vs. −0.305 for HML-Informed), indicating less severe tail breaches when they occur.

5.6 Robustness

Temporal dynamics. Figure 5 presents rolling 3-year Granger p -values for HML→SMB, showing that the predictive link is not a static property but concentrates during Crisis episodes. The signal is strongest during the 2008–2009 and 2020 crises and near-absent during calm periods.

Table 10: Hybrid Detector: COVID-19 Detection Sensitivity. Vol threshold = percentile of 20-day realized volatility computed on 1990–2012 training data. Primary result uses 95th percentile (bolded).

Percentile	Threshold	Override Days	Detection Rate
85th	0.924	85	92.4%
90th	1.098	76	82.6%
95th	1.446	44	47.8%
99th	2.122	25	27.2%

Table 11: Out-of-Sample VaR Backtest (2013–2024) with Hybrid Detector. Target violation rate: 5%.

Model	Viol. Rate	Dev. from 5%	CC p -value
Unconditional	6.42%	+1.42 pp	0.002
HML-Informed	5.77%	+0.77 pp	0.171
Hybrid (HMM+Vol)	5.60%	+0.60 pp	0.336

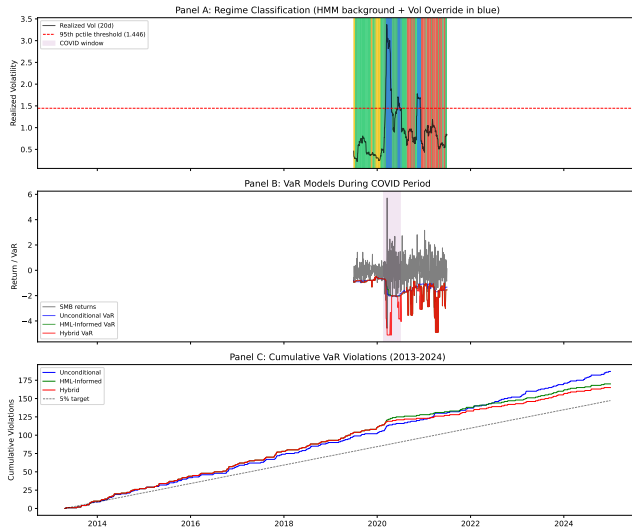


Figure 4: Hybrid detector performance. Panel A: realized volatility with 95th percentile threshold and override periods (COVID window highlighted). Panel B: SMB returns with unconditional, HML-informed, and hybrid VaR. Panel C: cumulative violations vs. 5% target.

Lag specification. BIC selects lag 1 for all regime $\times$ direction combinations (Table 3). Figure 6 shows significance is robust across lags 1–15 in Crisis, surviving at $\alpha = 0.001$.

Subsample stability. Splitting at 2008: Crisis HML $\rightarrow$ SMB holds strongly pre-2008 ($n = 571$ crisis days, $p = 1.0 \times 10^{-5}$) but weakens post-2008 ($n = 1,036$ crisis days, $p = 0.162$). This asymmetry may reflect changing market microstructure or evolving factor dynamics in the post-GFC environment.

Algorithm 1 Real-Time Factor Risk Monitor

Require: Frozen HMM parameters $(\mu_k, \Sigma_k, \nu_k, \mathbf{A})$; vol threshold σ_{95} ; HML threshold τ ; stress multiplier m ; regime windows (w_0, w_1, w_2)

- 1: **for** each trading day t **do**
- 2: Compute filtered regime: $\hat{z}_t \leftarrow \arg \max_k P(z_t = k \mid \mathbf{x}_{1:t})$
- 3: Compute 20-day realized vol: $\hat{\sigma}_t \leftarrow \text{std}(\|\mathbf{x}\|_{t-19:t})$
- 4: **if** $\hat{z}_t \neq \text{Crisis}$ **and** $\hat{\sigma}_t > \sigma_{95}$ **then**
- 5: $\hat{z}_t \leftarrow \text{Crisis}$ {Volatility override}
- 6: **end if**
- 7: Set VaR window: $w \leftarrow w_{\hat{z}_t}$
- 8: Compute base VaR: $\text{VaR}_t \leftarrow Q_{0.05}(\text{SMB}_{t-w:t-1})$
- 9: **if** $\hat{z}_t = \text{Crisis}$ **and** $\sum_{\ell=1}^9 \text{HML}_{t-\ell} < \tau$ **then**
- 10: $\text{VaR}_t \leftarrow m \cdot \text{VaR}_t$ {HML stress multiplier}
- 11: **end if**
- 12: **return** $\hat{z}_t, \text{VaR}_t$
- 13: **end for**

Table 12: Algorithm 1 Parameter Values. All calibrated on training data (1990–2012) only.

Parameter	Symbol	Value
Vol threshold (95th pctile)	σ_{95}	1.446
HML cumulative threshold	τ	−0.5
Stress multiplier	m	2.3
Window (Calm)	w_0	75 days
Window (Normal)	w_1	50 days
Window (Crisis)	w_2	30 days
HML lookback	—	9 days

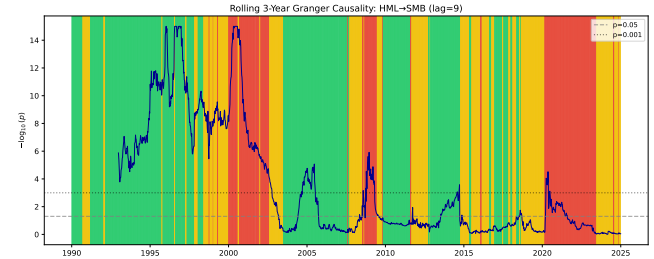


Figure 5: Rolling 3-year Granger causality ($-\log_{10}(p)$, HML $\rightarrow$ SMB, lag=9). Background shading shows regime assignments. Predictive strength concentrates during Crisis periods and is not a static feature of the data.

Alternative regime methods. Threshold-based volatility regimes (realized volatility $>$ 90th percentile = Crisis) also detect the pattern ($p = 0.0003$, lag 7 vs. 9), confirming it is not HMM-specific.

Practitioner baselines. Standard unconditional tools detect the HML $\rightarrow$ SMB asymmetry but miss the architectural distinction. Rolling 250-day Granger tests find HML $\rightarrow$ SMB significant in 36.2% of windows vs. 9.6% for the reverse—but

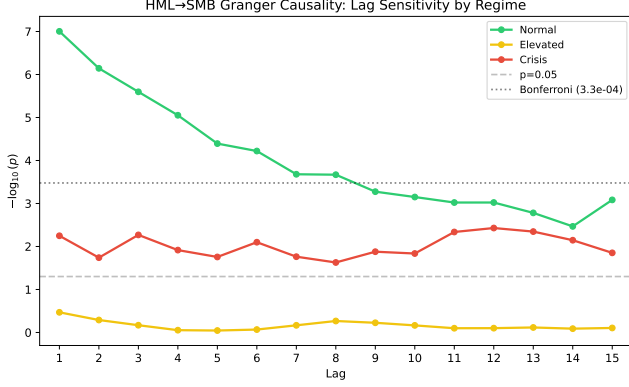


Figure 6: Granger causality p -values for HML→SMB across lags 1–15 by regime. Normal regime (green) shows concentrated short-lag significance ($p < 10^{-7}$ at lag 1). Crisis regime (red) shows a broad plateau ($p < 0.025$ across all lags), indicating persistent multi-lag predictive structure.

significance concentrates in Normal (47.7%) and is *lowest* in Crisis (17.6%), because regime mixing dilutes the multi-lag persistence that our approach isolates. FEVD spillover is regime-uniform (7.7–10.2%), failing to detect the dormant Elevated regime or the Normal/Crisis directionality shift. Regime conditioning does not merely sharpen detection—it reveals qualitatively different predictive architectures that rolling methods average away.

Weekly aggregation. Using weekly returns, the HML→SMB crisis pattern loses significance ($p = 0.554$), likely due to reduced sample size (crisis regime: 371 weeks).

Regime transitions. Analyzing 60-day windows around Normal→Crisis transitions, the HML→SMB relationship strengthens upon crisis entry (median p : 0.18 → 0.03), suggesting discrete intensification of the predictive link.

Model complexity diagnostic protocol. We apply four model classes—Linear (OLS), RF (100 trees), MLP (64-32), LSTM (32 hidden, 9-step)—to characterize the *complexity boundary* of the predictive relationship. Each predicts SMB from its own lags (restricted) vs. both SMB and HML lags (unrestricted), with permutation testing (200 shuffles for Linear/RF/MLP, 100 for LSTM; at 200 permutations, p -value SE ≈ 0.007 at $p = 0.01$) and expanding-window CV.

Table 13 and Figure 7 present the results. The linear Granger effect shows a clear regime gradient (Crisis 1.25% > Normal 0.73% > Elevated 0.30% MSE improvement). However, *no nonlinear method improves prediction in any regime*: RF ($p = 0.96$), MLP ($p = 0.93$), and LSTM ($p = 0.83$) all fail to improve upon the linear baseline during Crisis.² Gradient-based feature importance reveals a telling dissociation: the LSTM assigns 49% of input importance to HML features during Crisis (comparable to 58% during Normal), confirming

²The LSTM was estimated under a separate HMM initialization (see footnote 1); qualitative conclusions are consistent across fits.

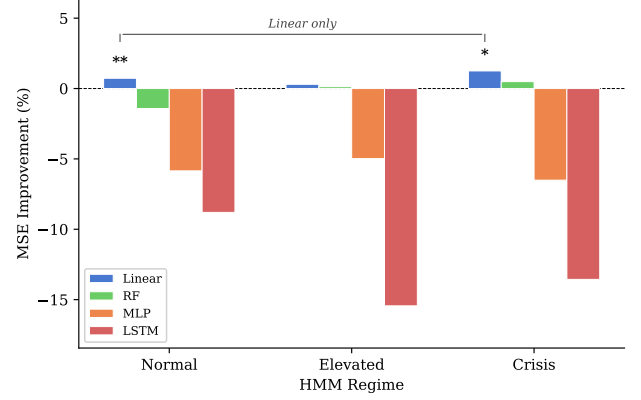


Figure 7: Four-model diagnostic protocol: MSE improvement (%) from including HML lags, by regime and model class. Only linear models (blue) achieve significant improvement in Normal and Crisis; all nonlinear methods (RF, MLP, LSTM) fail to improve prediction. * $p < 0.05$; ** $p < 0.001$.

Table 13: Four-Model Diagnostic Protocol: HML→SMB by Regime. Linear column shows MSE improvement (%); others show permutation p -values.

Regime	n	Linear	RF p	MLP p	LSTM p
Normal	4,049	0.73%**	1.00	0.64	0.63 [†]
Elevated	2,520	0.30%	0.96	0.91	0.93 [†]
Crisis	1,675	1.25%*	0.96	0.93	0.83 [†]

** $p < 0.001$, * $p < 0.05$. [†]LSTM: separate HMM init., 100 perms.

that the network *detects* HML relevance but *cannot leverage* it beyond linear prediction. This complexity characterization is the key methodological finding: the predictive link is structurally linear, operating through mean dynamics rather than nonlinear channels.

Transfer entropy confirmation. As a model-free check, we compute transfer entropy [18] (Frenzel–Pompe kNN, $k = 5$, 200 permutations). In Elevated and Crisis, HML→SMB TE is significant (Elevated: $z = 2.89$, $p = 0.002$; Crisis: $z = 2.37$, $p = 0.009$) while the reverse is not ($p = 0.058$; $p = 0.188$), confirming unidirectional flow. Normal shows significant bidirectional TE (SMB→HML: $z = 4.08$, $p < 0.001$), despite non-significant linear Granger ($p = 0.49$, Table 3) and multi-lag ΔR^2 ($p = 0.295$, Table 4). This discrepancy suggests a *nonlinear* reverse-direction channel during calm conditions, reinforcing our main finding: during Crisis, both linear and nonparametric methods agree on unidirectional HML→SMB flow.

Filtered vs. smoothed regime probabilities. Smoothed probabilities use the full sample (forward and backward pass); filtered probabilities use only past data ($P(z_t = k \mid$

Table 14: Transfer Entropy: Multi-Lag (1–9) by Regime. z -scores relative to permutation null (200 shuffles).

Regime	n	HML→SMB z	p	SMB→HML z	p
Normal	4,049	2.33*	0.010	4.08***	< 0.001
Elevated	2,520	2.89**	0.002	1.57	0.058
Crisis	1,675	2.37**	0.009	0.89	0.188

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Elevated and Crisis: unidirectional HML→SMB; Normal: bidirectional.
 n = clean observations after boundary exclusion (lags 1–9 within regime).

$x_1, \dots, x_t$), simulating real-time classification. Overall agreement is 93.9%. Under filtered regimes, the Crisis HML→SMB result yields $p = 0.007$ (vs. $p = 5.6 \times 10^{-3}$ smoothed), with filtered Crisis containing 1,840 days (vs. 1,807 smoothed). The finding is qualitatively consistent across classification methods.

5.7 Limitations

- (1) **Granger $\neq$ structural causality.** Our core finding is that HML *Granger-causes* SMB during crises—i.e., lagged HML improves SMB prediction conditional on lagged SMB. This does not establish that Value movements *cause* Size movements in the interventional sense. A common factor (liquidity, sentiment) could drive both with different lags. The “deleveraging cascade” interpretation is a plausible hypothesis, not a verified mechanism.
- (2) **Portfolio-level (not holdings-level) mechanism evidence.** Our FF 25 overlap analysis (Section 4.5) provides portfolio-level evidence for the deleveraging channel, but we do not analyze 13F holdings data or fund flows for direct verification.
- (3) **Sample overlap in regime identification.** Regime labels use full-sample information (Section 3.3). Our frozen-parameter validation (Section 5.2) shows that strictly out-of-sample regime classification does not reproduce the aggregate Crisis-level result ($p = 0.138$), though the 2020 COVID event-specific pattern persists ($p = 0.032$).
- (4) **Regime boundary exclusion and sample size.** Boundary cleaning (1–15 lags excluded per table) reduces effective samples. Crisis contains 1,807 days; the Crisis result ($p = 5.6 \times 10^{-3}$) does not survive Bonferroni correction.
- (5) **Normal > Crisis and subsample instability.** Normal HML→SMB ($p = 9.9 \times 10^{-8}$) exceeds Crisis, and the Crisis effect is driven by pre-2008 data ($p = 1.0 \times 10^{-5}$; post-2008: $p = 0.162$). The regime-dependent finding concerns intensity (multi-lag persistence), not presence.
- (6) **Risk monitoring scope.** VaR improvement is modest (6.42%→5.60% with hybrid detector) and may partly reflect conservative widening during high-volatility regimes

rather than the Granger link itself. The hybrid detector’s volatility override addresses the regime classifier’s COVID-19 failure but adds one additional parameter (95th percentile threshold).

6 Conclusion

We document that the HML→SMB predictive link has qualitatively distinct architecture across market regimes. In Normal conditions, the effect is concentrated at lag 1 ($p < 10^{-7}$) with bidirectional nonlinear channels detectable by transfer entropy. In Crisis, it extends across lags 1–15 ($\Delta R^2 = 1.83\%$) but becomes strictly unidirectional under both linear and non-parametric tests. This architectural distinction—not merely a difference in magnitude—suggests different economic mechanisms: rapid information incorporation during calm markets versus gradual deleveraging during stress.

Does establish:

- Qualitatively distinct predictive architectures across regimes, confirmed by both two-stage (HMM + Granger) and single-stage (MS-VAR) approaches, robust to HAC correction, structurally linear
- Portfolio-level mechanism evidence: Granger significance concentrates in small-cap portfolios ($\rho_s = 0.35$, permutation $p = 0.046$), with the Small-HighBM portfolio accounting for 39% of the effect
- A TE/Granger discrepancy revealing nonlinear reverse-direction channels in Normal that are invisible to standard linear tests
- Temporal precedence (HML leads SMB) across regimes

Does not establish:

- Robust out-of-sample regime-level significance (frozen HMM: $p = 0.138$)
- Structural causality (common drivers could explain the pattern)
- Trading profitability ($\Delta R^2 \approx 2\%$; Sharpe = -0.56)

Practical value: A hybrid VaR model combining HMM regime detection with a realized-volatility fallback achieves the closest coverage to the 5% target (5.60%, $p_{CC} = 0.336$), detecting 47.8% of COVID-2020 where the HMM alone detected 0%. Algorithm 1 formalizes the monitoring protocol with all parameters calibrated on training data.

For practitioners, the key insight is that factor lead-lag relationships are not merely stronger or weaker across regimes—they are *structurally different*. This distinction translates into concrete monitoring rules: during Normal regimes, a 1-day HML lookback suffices for SMB risk, but nonlinear reverse-direction channels (SMB→HML) should also be monitored via TE-based indicators; during Crisis regimes, extend HML monitoring to a 15-day lookback to capture the multi-lag deleveraging signal, while reverse-direction monitoring can be deprioritized. Standard rolling Granger tests invert this regime ordering (Section 5.6), making regime conditioning essential for correct monitoring architecture. We note that the Crisis-regime architecture is empirically validated for liquidity-driven crises with factor crowding (2008, 2020);

policy-driven episodes (2018, 2022) show different dynamics. Real-time implementation previously faced a regime-detection latency constraint: the frozen HMM classified 0% of COVID-2020 as Crisis (Table 7). The hybrid detector (Algorithm 1) resolves this by supplementing HMM-based regime detection with a realized-volatility override, detecting 47.8% of the COVID period while adding only one pre-specified parameter (95th percentile of training-period volatility). TE-based reverse-direction monitoring can be approximated using rolling 250-day windows ($k = 5$ nearest neighbors) updated weekly against a pre-computed null distribution.

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